Two Tangents from a Point

Exam code: 0607

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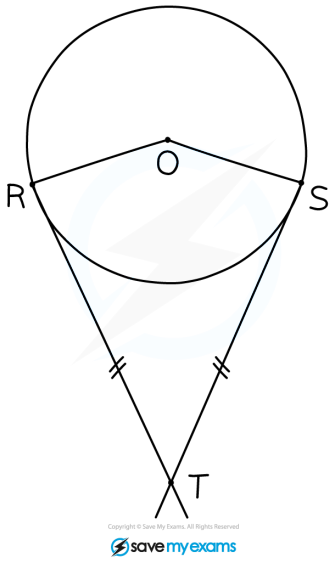
Updated on 26 June 2024

Two Tangents from a Point

Circle Theorem: Tangents from an external point are equal in length

- **Two tangents** from the **same external point** are equal in length
- This means that a **kite** can be formed by two tangents meeting a circle
  - The kite below has a vertical line of symmetry
    - It is formed from two **congruent** triangles back-to-back
  - The kite will have **two right angles** where the tangents meet the radii
    - You can use **Pythagoras** and **SOHCAHTOA** on each of these triangles



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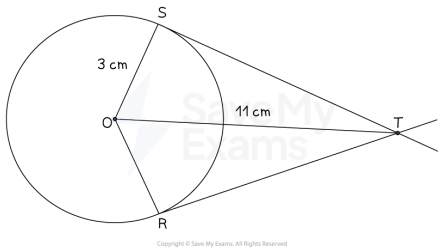
Worked Example

Consider the diagram below.



(a) Find the length *ST*.

The lines *ST* and *RT* are both tangents to the circle
They meet the two radii on the circumference at the points *S* and *T*

Angle *TSO* = 90° because a radius and a tangent meet at right angles

Use Pythagoras' theorem on the right-angled triangle *OST* to calculate the length *ST*

$$3^2 + (ST)^2 = 11^2$$
$$9 + (ST)^2 = 121$$
$$(ST)^2 = 112$$
$$ST = \sqrt{112} = 10.58300...$$

ST = 10.6 cm (3 s.f.)

(b) Find the length *RT*.

Two tangents from the **same external point** are equal in length

$$ST = RT$$

RT = 10.6 cm (3 s.f.)

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